

Text to Speech?

All the fancy GUIs aside, computers are still, primarily, tools for manipulating text. Most current office and Internet usage is plain text-e-mail, Web pages, chat sessions and documents. Text is no longer the primary way to interact with a PC, but manipulating text is still the primary application for PCs.

The name 'computer' underscores its origin as a pure number crunching machine. Computers would really become useful to those of us non-mathematically inclined when text-capture hardware, or keyboards, became a common and affordable add-on. Keyboards, for all practical purposes, remain indispensable 'human interface devices'. Humans are, of course, far more comfortable working with visual and audible representations of information than with typing and reading.

Now, two fundamental shifts are on the verge—images have an affordable, convenient channel into the PC, thanks to the unstoppable rise of digital cameras. Audio recording has been feasible for a while, but until recently, computers simply didn't have the muscle to transform and wield audio in any meaningful way. More importantly, as communication devices proliferate, we are approaching the bandwidth to move acceptable quality audio and video around almost as conveniently as text—witness the explosive growth of Internet telephony combined with the Web cam.

We naturally interact better with audio and visual cues than with creating and using text. As it becomes easier to channel audio, video and images through the PC, it becomes imperative that we have the same level of control over it as we currently do over text.

When you have a terabyte of 'folders' filled with conversation, packaged communications, presentations and video-mails on your 400 GB hard disk, how do you find what you need, repurpose it for other people or situations, reformat it and so on? Where is the Word for video documents? Laptops have audio/visual recording capabilities. Why, for example, are minutes of meetings not recorded? Where is the indexing service for spoken language? How do you find the meeting note where your priorities were discussed?

Communicating out of such a repository is a challenge. Finding interlinks between all that information is a challenge. Simply finding the bit you want out of it is a challenge. Many of these challenges could be fixed by simply throwing a bucketful of CPUs at them. We can't find what we want instantly on our own PCs, because running the software to do that imposes a resource overhead—and that overhead we would currently classify only as luxury.

Processing power needs to significantly outstrip what we have at hand today before this digital age can be real, and it will. To paraphrase James Gosling, the acknowledged father of the Java programming language, "If Moore's Law holds true, and you have a problem that will take current computers a 1,000 years to solve, you might as well go away to a beach for 14 years, come back and solve the problem in one year."

Time to hit that beach.



sumod_hajela@jasubhai.com



Sumod Hajela
Assistant Editor

“Many of these challenges could be fixed by simply throwing a bucketful of CPUs at them.”